

624 Switching Power Supply Design

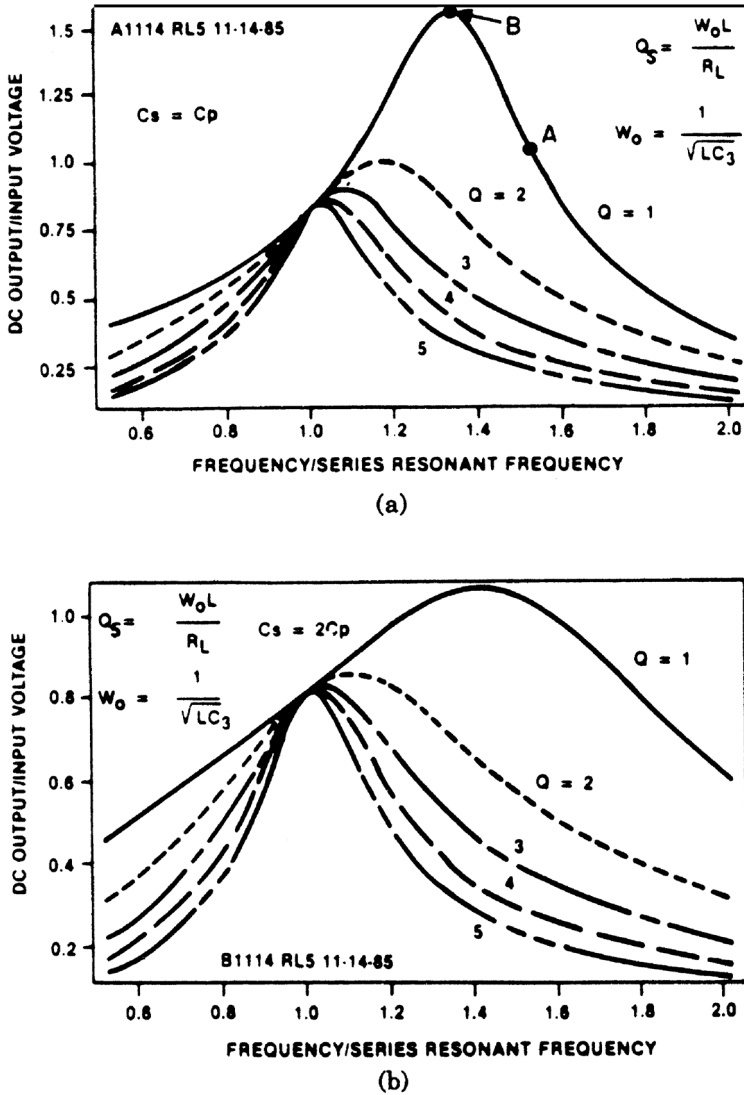


FIGURE 13.8 (a) Gain curves for series-parallel half-bridge resonant converter (LCC) of Figure 13.5c for $C_s = C_p$. (b) Gain curves for series-parallel half-bridge resonant converter (LCC) of Figure 13.5c for $C_s = 2C_p$. (Courtesy of R. Steigerwald.)

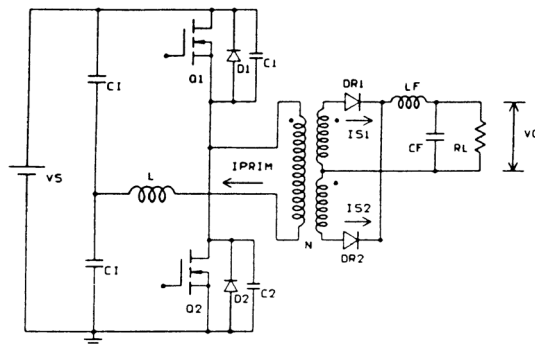


FIGURE 13.9 A zero-voltage-switching half-bridge resonant converter.
(Courtesy of Jovanovic, Tabisz, Lee.)

zero at the end of the first half period (or shortly after when current has reversed and flows into the antiparallel diode as in Figure 13.1), there is no overlap of high voltage and current and turn “off” losses are eliminated.

However, there are turn “on” losses even though there is no overlap of high voltage and current because the leakage inductance speeds up drain (or collector) voltage fall time and slows up drain current rise time. These losses occur because the drain capacitance stores an energy at turn “off” of $0.5C(V_{\max})^2$, and dissipates it in the transistor at the next turn “on.” This happens once per cycle and results in average dissipation of $0.5C(V_{\max})^2/T$, which becomes significant over 500 kHz to 1 MHz. As designers move to these higher frequencies to reduce size, this becomes a problem.

A new technique—ZVS—has been proposed to circumvent this problem. It has been proposed for single-ended (flyback or buck) circuits¹⁰, but its value is mainly for half bridges.

A ZVS half bridge is shown in Figure 13.9.² Its basic principle is that the MOSFET output capacitor is used as part of the capacitor of the resonant LC circuit. It stores a voltage and hence energy in one part of the switching cycle. In a following part of the switching cycle, the energy in the capacitor is discharged through the resonant inductor back into the supply bus without dissipation. The following discussion is based on Reference 2.

In Figure 13.9, when Q1 is “on” and Q2 is “off,” C2 is charged up to V_s . Transistor Q1 is first turned “off.” Transformer T1 magnetizing current continues to flow through C1, pulling its voltage down. Halfway down, there is no voltage across the primary and in the T1 secondary, the output inductor tries to maintain a constant current.

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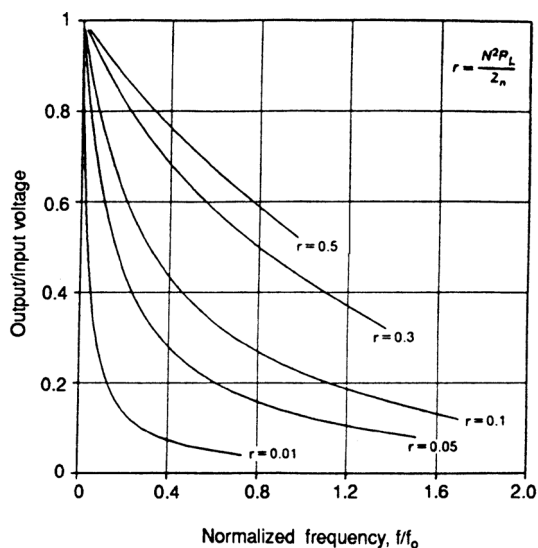


FIGURE 13.10 Gain curve for the zero-voltage-switching circuit of Figure 13.9. (Courtesy of Jovanovic, Tabisz, Lee.)

Both rectifier diodes turn “on,” which provides a short circuit across the secondary and hence across the primary.

At this point, the energy stored on the top end of C_2 discharges through the short-circuited primary, through the resonant L , through the bottom filter capacitor C_{r2} , and back into the bottom end of C_2 . Since there are no resistors in this path, the discharge is lossless. The negative resonant voltage impulse on the right hand of L pulls the junction of C_1 , C_2 down to ground, and now Q_2 is turned “on” at zero voltage.

Capacitor C_1 has slowed up Q_1 voltage fall time sufficiently, so that there is no simultaneous high voltage and current during its turn “off.”

The circuit, a CCM type, operates on the slope of the resonant curve of the circuit consisting of L and C_1 , C_2 in parallel. Jovanovic et al. give the DC voltage conversion curve for the circuit (the equivalent of Figures 13.5 to 13.8) in Figure 13.10.²

Despite the many articles written on CCM, which is regulated by changing frequency along the steep slope of a resonant curve, this author feels it is not a reliable scheme. It will not yield similar results over a large production run of supplies with all the various production spreads in the resonant components and with the limited range of V_{dc} and R_l in the conversion curves such as those in Figures 13.5 to 13.8 and 13.10.

13.6 Resonant Power Supplies—Conclusion

To the engineer who must decide whether to consider a resonant supply for the next design, the following should be considered:

1. Can I get the same (or acceptable) power density in watts per cubic inch with a conventional PWM supply operating at, say, 200 to 300 kHz? Is the added complexity, limited line and load capability, and difficulty to design for worst-case conditions worth the extra 3 to 6% better efficiency possible with resonant supplies?
2. Will all units coming off a production line have identical characteristics? Or because of tolerances and production spread in component values, will fine-tuning of each unit or a large fraction of the production run be required?
3. Do resonant supplies actually generate less RFI (because their currents are sinusoidal rather than square wave) after you factor in their higher di/dt ? Consider that most resonant supplies have sine wave currents three to four times the amplitude of PWM square-wave supplies of equal power. It is possible that the RFI problem with resonant supplies may be as severe since di/dt at the zero crossing of a sine wave is proportional to its peak value.
4. How serious is the problem that most resonant supplies regulate by varying the frequency? Will users accept this or insist that the switching supply be synchronized to a clock signal which they will supply?
5. If the decision is to attempt a resonant design, which of the bewildering number of topologies advocated would be the safest approach? Are continuous-mode designs too unpredictable because they require operation on the slippery slope of a narrow resonant curve for regulation? There seems to be general agreement that discontinuous-mode operation is more predictable and reproducible. Is the larger frequency range required in discontinuous mode an important drawback?
6. There is no question that high-frequency resonant converters will continue to be studied and improved. Until configurations are found which lend themselves to simple, worst-case designing and which are as insensitive as PWM circuits to component tolerances, wiring layout, and parasitic inductances and capacitances, high-frequency resonant converters will not be widely adopted—certainly not in programs with a large production run. They will occupy only a narrow niche in the power supply field where higher cost and finely tuned component selection and wiring layout are acceptable.

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In summary, in any design including switching power supplies, complexity equals higher cost and possible unreliability.

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